

Lean port sheets

Rev F0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | D-044 lean ports A (PV) and B (battery) + insulation monitor, BUS- referenced

01 Supplies and commands

Live 24 V / 5 V / 3.3 V, BUS- tie, commands

File: sheets/01_lean_io.kicad_sch

02 Port A power path (pv)

Varistor Y + monitor

File: sheets/02_leanA_power.kicad_sch

03 Port A sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/03_leanA_ctrl.kicad_sch

04 Port A coil drives

Contactor: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/04_leanA_drive.kicad_sch

05 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/05_leanB_power.kicad_sch

06 Port B sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/06_leanB_ctrl.kicad_sch

07 Port B coil drives

Contactor + precharge relay: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/07_leanB_drive.kicad_sch

08 Insulation monitor

PE divider, two switched 992 k strings through CA-IS3417WT

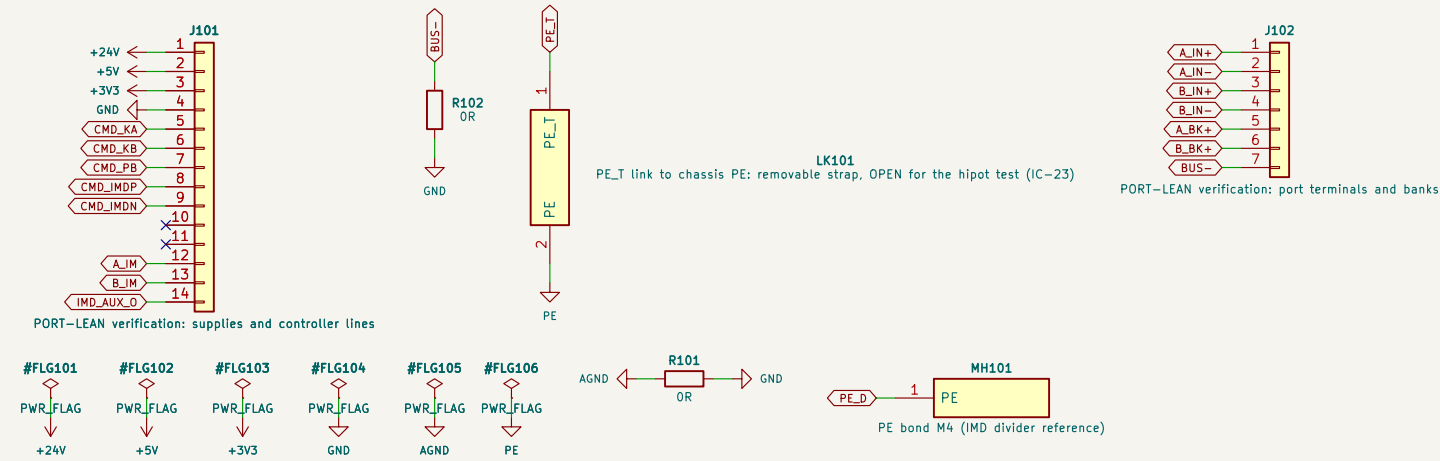
File: sheets/08_lean_imd.kicad_sch

Verification build of the reusable lean-port functions of gen/port.py (lean_port, varistor_y, lean_contactor, lean_precharge, lean_bleeder, lean_dividers, lean_shunt, lean_interlock, lean_imd). The product sheets live on the power board.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: / File: PORT-LEAN-HOLD.kicad_sch		
Title: Lean port sheets – Cover / Hierarchy		
Size: A3	Date: 2026-10-04	Rev: F0
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SUPPLIES AND COMMANDS



On the power board these come from the AUX-T1 live winding and the controller.

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Sheet 2
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /01 Supplies and commands/
File: 01_lean_io.kicad_sch

Title: Lean port sheets – Supplies and commands

Size: A3	Date: 2026-10-04	Rev: F0
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PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{LN}+ / A_{LN}-, bank A_{BK}+, No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above [port_spec lean]. Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN+} / A_{IN-}, bank A_{BK+}. No fuse, no precharge: array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up,eff 3793 V pole-PE and pole-pole at in 3 kA (calc., ≤ 4.0 kV $\rightarrow$ reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 k (L/R 2 ms). Monitor loop through both fuses and all three links $\rightarrow$ CNY65B $\rightarrow$ RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open $\rightarrow$ CTRL inhibits/stops.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

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Sheet 4

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /03 Port A sensing and interlocks/

File: 03_leanA_ctrl.kicad_sch

Title: Lean port sheets – Port A sensing and interlocks

Size: A3

Date: 2026-10-04

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	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'		
Sheet 5		
Schematics authored by Chinmoy Bhuyan		
Live side referenced to BUS-; basic HV-PE; functional inside the live side		
Sheet: /04 Port A coil drives/ File: 04_leanA_drive.kicad_sch		
Title: Lean port sheets – Port A coil drives		
Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6	Id: 5/9	

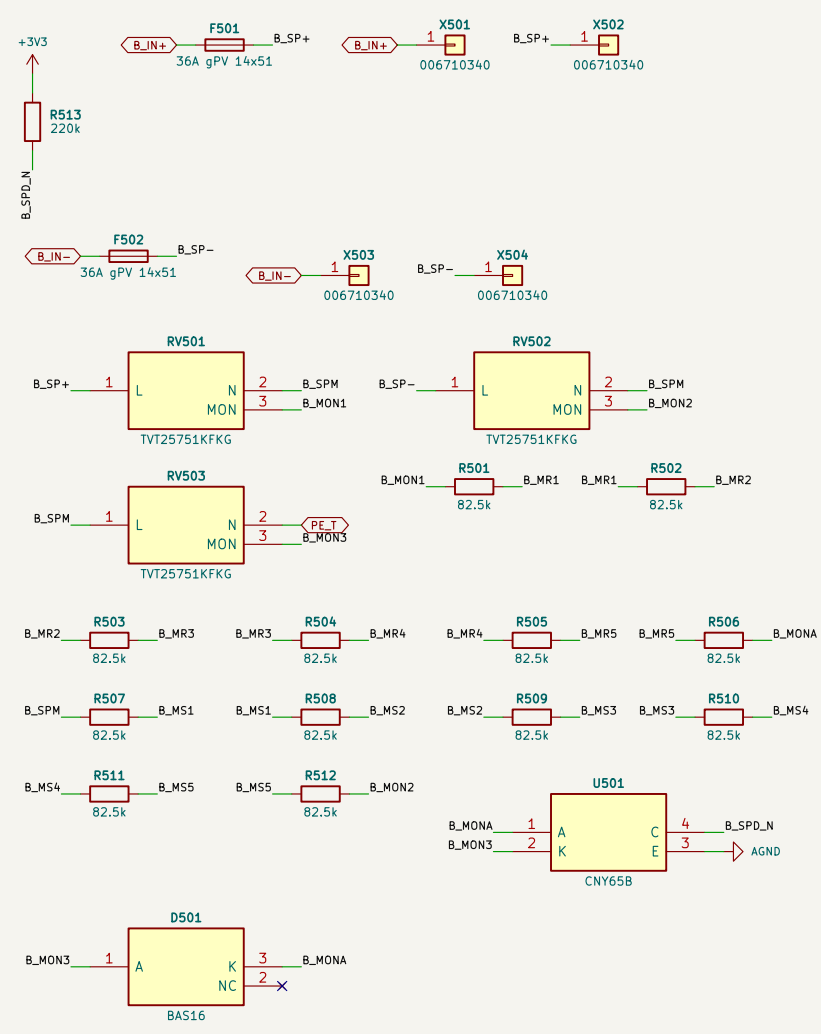
PORT B (BATTERY SOURCE), D-044 LEAN PORT

Terminal side B₁IN+ / B₁IN-, bank B₁BK+. HPE501 250 A aR in both poles (breaks 1.25–50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

Diagram illustrating the on-board surge protection circuit for Varistor Y (IC-01, IC-12). The circuit includes a varistor (RV501) connected to the +V5V supply and ground, and a network of resistors (R501-R506) and capacitors (C501, C502) connected to the +V5V supply and ground. The circuit is designed to protect the system from voltage spikes and surges.

Component Values:

- RV501: 36A 5pV 14x51
- R501: 36A 5pV 14x51
- R502: 36A 5pV 14x51
- R503: 82.5k
- R504: 82.5k
- R505: 82.5k
- R506: 82.5k
- R507: 82.5k
- R508: 82.5k
- R509: 82.5k
- R510: 82.5k
- R511: 82.5k
- R512: 82.5k
- R513: 82.5k
- R514: 82.5k
- R515: 82.5k
- R516: 82.5k
- R517: 82.5k
- R518: 82.5k
- R519: 82.5k
- R520: 82.5k
- R521: 82.5k
- R522: 82.5k
- R523: 82.5k
- R524: 82.5k
- R525: 82.5k
- R526: 82.5k
- R527: 82.5k
- R528: 82.5k
- R529: 82.5k
- R530: 82.5k
- R531: 82.5k
- R532: 82.5k
- R533: 82.5k
- R534: 82.5k
- R535: 82.5k
- R536: 82.5k
- R537: 82.5k
- R538: 82.5k
- R539: 82.5k
- R540: 82.5k
- R541: 82.5k
- R542: 82.5k
- R543: 82.5k
- R544: 82.5k
- R545: 82.5k
- R546: 82.5k
- R547: 82.5k
- R548: 82.5k
- R549: 82.5k
- R550: 82.5k
- R551: 82.5k
- R552: 82.5k
- R553: 82.5k
- R554: 82.5k
- R555: 82.5k
- R556: 82.5k
- R557: 82.5k
- R558: 82.5k
- R559: 82.5k
- R560: 82.5k
- R561: 82.5k
- R562: 82.5k
- R563: 82.5k
- R564: 82.5k
- R565: 82.5k
- R566: 82.5k
- R567: 82.5k
- R568: 82.5k
- R569: 82.5k
- R570: 82.5k
- R571: 82.5k
- R572: 82.5k
- R573: 82.5k
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- R576: 82.5k
- R577: 82.5k
- R578: 82.5k
- R579: 82.5k
- R580: 82.5k
- R581: 82.5k
- R582: 82.5k
- R583: 82.5k
- R584: 82.5k
- R585: 82.5k
- R586: 82.5k
- R587: 82.5k
- R588: 82.5k
- R589: 82.5k
- R590: 82.5k
- R591: 82.5k
- R592: 82.5k
- R593: 82.5k
- R594: 82.5k
- R595: 82.5k
- R596: 82.5k
- R597: 82.5k
- R598: 82.5k
- R599: 82.5k
- R600: 82.5k
- R601: 82.5k
- R602: 82.5k
- R603: 82.5k
- R604: 82.5k
- R605: 82.5k
- R606: 82.5k
- R607: 82.5k
- R608: 82.5k
- R609: 82.5k
- R610: 82.5k
- R611: 82.5k
- R612: 82.5k
- R613: 82.5k
- R614: 82.5k
- R615: 82.5k
- R616: 82.5k
- R617: 82.5k
- R618: 82.5k
- R619: 82.5k
- R620: 82.5k
- R621: 82.5k
- R622: 82.5k
- R623: 82.5k
- R624: 82.5k
- R625: 82.5k
- R626: 82.5k
- R627: 82.5k
- R628: 82.5k
- R629: 82.5k
- R630: 82.5k
- R631: 82.5k
- R632: 82.5k
- R633: 82.5k
- R634: 82.5k
- R635: 82.5k
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- R640: 82.5k
- R641: 82.5k
- R642: 82.5k
- R643: 82.5k
- R644: 82.5k
- R645: 82.5k
- R646: 82.5k
- R647: 82.5k
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- R653: 82.5k
- R654: 82.5k
- R655: 82.5k
- R656: 82.5k
- R657: 82.5k
- R658: 82.5k
- R659: 82.5k
- R660: 82.5k
- R661: 82.5k
- R662: 82.5k
- R663: 82.5k
- R664: 82.5k
- R665: 82.5k
- R666: 82.5k
- R667: 82.5k
- R668: 82.5k
- R669: 82.5k
- R670: 82.5k
- R671: 82.5k
- R672: 82.5k
- R673: 82.5k
- R674: 82.5k
- R675: 82.5k
- R676: 82.5k
- R677: 82.5k
- R678: 82.5k
- R679: 82.5k
- R680: 82.5k
- R681: 82.5k
- R682: 82.5k
- R683: 82.5k
- R684: 82.5k
- R685: 82.5k
- R686: 82.5k
- R687: 82.5k
- R688: 82.5k
- R689: 82.5k
- R690: 82.5k
- R691: 82.5k
- R692: 82.5k
- R693: 82.5k
- R694: 82.5k
- R695: 82.5k
- R696: 82.5k
- R697: 82.5k
- R698: 82.5k
- R699: 82.5k
- R700: 82.5k
- R701: 82.5k
- R702: 82.5k
- R703: 82.5k
- R704: 82.5k
- R705: 82.5k
- R706: 82.5k
- R707: 82.5k
- R708: 82.5k
- R709: 82.5k
- R710: 82.5k
- R711: 82.5k
- R712: 82.5k
- R713: 82.5k
- R714: 82.5k
- R715: 82.5k
- R716: 82.5k
- R717: 82.5k
- R718: 82.5k
- R719: 82.5k
- R720: 82.5k
- R721: 82.5k
- R722: 82.5k
- R723: 82.5k
- R724: 82.5k
- R725: 82.5k
- R726: 82.5k
- R727: 82.5k
- R728: 82.5k
- R729: 82.5k
- R730: 82.5k
- R731: 82.5k
- R732: 82.5k
- R733: 82.5k
- R734: 82.5k
- R735: 82.5k
- R736: 82.5k
- R737: 82.5k
- R738: 82.5k
- R739: 82.5k
- R740: 82.5k
- R741: 82.5k
- R742: 82.5k
- R743: 82.5k
- R744: 82.5k



TV125751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/M, MOV2 -/M, MOV3 M/PE,T; Up,eff 3793 V pole-PE and pole-pole at in 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); Imax 25 kA, Branch

USES (BATTERY-TYPE PORT)

HPE501/010B100-250 in both poles: breaks 1.25-50 kA only (partial range)

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 7
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /06 Port B sensing and interlocks/
File: 06_leanB_ctrl.kicad_sch

Title: Lean port sheets – Port B sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

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Id: 7/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 8

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /07 Port B coil drives/

File: 07_leanB_drive.kicad_sch

Title: Lean port sheets – Port B coil drives

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 8/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 9

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /08 Insulation monitor/

File: 08_lean_imd.kicad_sch

Title: Lean port sheets – Insulation monitor

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 9/9